

Step 2: Incorporate the forecasted volatility from the preferred forecasting model, e.g., HMA-VIX, GARCH, EWMA, etc., into the new covariance matrix $\hat{C}$:

$$\hat{C} = \hat{D}(\text{Rho})\hat{D} = \hat{D}\hat{D}^{-1}CD^{-1}\hat{D}$$

This covariance matrix will now be scaled to the same time period as the price volatility term. For example, if the volatility forecast is a one-day forecast then the covariance matrix $\hat{C}$ is also a one-day estimate. If we are interested in a time period that is different than the time scale of the price volatility estimate we simply divide by that appropriate value. For example, if we break the day into n trading periods the covariance matrix for the time horizon is:

$$\hat{C} = \frac{1}{n} \cdot \hat{D}(\text{Rho})\hat{D} = \frac{1}{n} \cdot \hat{D}\hat{D}^{-1}CD^{-1}\hat{D}$$

Step 3: Convert the covariance matrix expressed in (returns)² into (\$/share)². Here we simply multiply by our diagonal price matrix from P above:

$$\tilde{C} = P\hat{C}P = \frac{1}{n} \cdot P\hat{D}\hat{D}^{-1}CD^{-1}\hat{D}P$$

This covariance matrix is now scaled for the appropriate length of time for our trading period and is expressed in (\$/share)² for our trade schedule timing risk calculations. This matrix will also be extremely important for portfolio optimization.

The general form of our trading risk model is:

$$\tilde{C} = \frac{1}{n} \cdot P\hat{D}\hat{D}^{-1}CD^{-1}\hat{D}P \quad (7.55a)$$

Many times investors will need the covariance matrix to be adjusted for a one-sided portfolio. In this case, we adjust the entries in the covariance matrix based on the side of the order. For example,

$$c_{ij}^* = \text{side}(i) \cdot \text{side}(j) \cdot c_{ij} \quad (7.55b)$$

Where c_{ij} is the computed covariance scaled for the length of the trading period and in (\$/share)². The full side adjusted covariance is computed via matrix multiplication following techniques above as follows:

$$\tilde{C} = \frac{1}{n} \cdot (\text{Side})P\hat{D}\hat{D}^{-1}CD^{-1}\hat{D}P(\text{Side}) \quad (7.55c)$$

Where (*Side*) is a diagonal matrix consisting of either a 1 if a buy order or -1 if a sell order. We make use of the side adjusted trading risk covariance in Chapter 8 through Chapter 10.